SC 05 · QUADROTOR SIMULATION

From one axis to six degrees of freedom

The equations, the mixing matrix, and the cascade, on the 5-inch quad of the rotorfault study.

Rigid body in one screen

$$m\dot{v} = R\,(0,0,T)^T - RDR^Tv - mg\,e_3, \qquad J\dot{\omega} = \tau - \omega \times J\omega - K_\omega\omega$$

- thrust only along body z: **tilt to move**
- rotor: $f_i = k_f\Omega_i^2$, reaction torque $s_i\kappa f_i$, lag $\tau_m = 25$ ms
- `bootcamp.quad` : ~300 lines, RK4, every PID is easyPID

The allocation matrix

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -a & a & a & -a \\ -a & -a & a & a \\ \kappa & -\kappa & \kappa & -\kappa \end{bmatrix} f$$

Square and invertible: four motors, four quantities, **no spare actuator**.

Hover: $f_i = mg/4, \Omega_h = \sqrt{mg/4k_f} \approx 1321$ rad/s, thrust-to-weight 3.6.

When the motors saturate

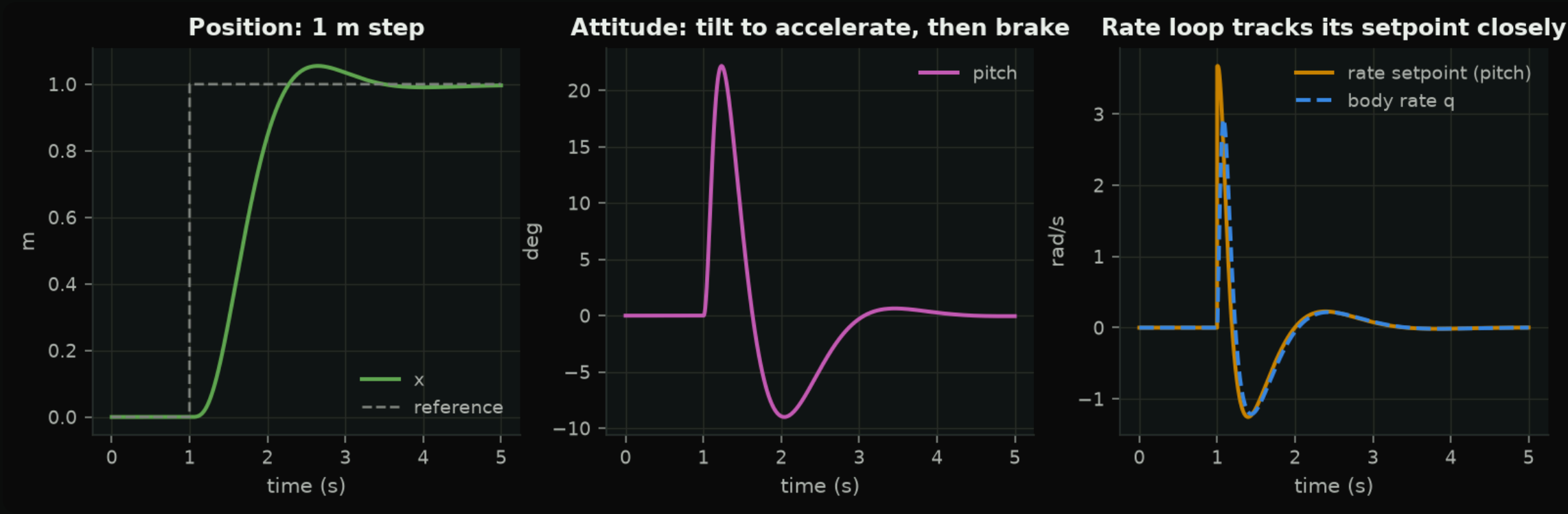
1. keep **roll/pitch** (it keeps you in the air)
2. shift **collective thrust** (air mode)
3. scale **yaw** first: small authority (κ = 1.5 cm vs a 7.8 cm lever), harmless to lose

Acceleration → tilt

$$\theta_d = \frac{a_x \cos\psi + a_y \sin\psi}{g}, \qquad \phi_d = \frac{a_x \sin\psi - a_y \cos\psi}{g}$$

$$T = \frac{m(g + a_z)}{\cos\phi \cos\theta}$$

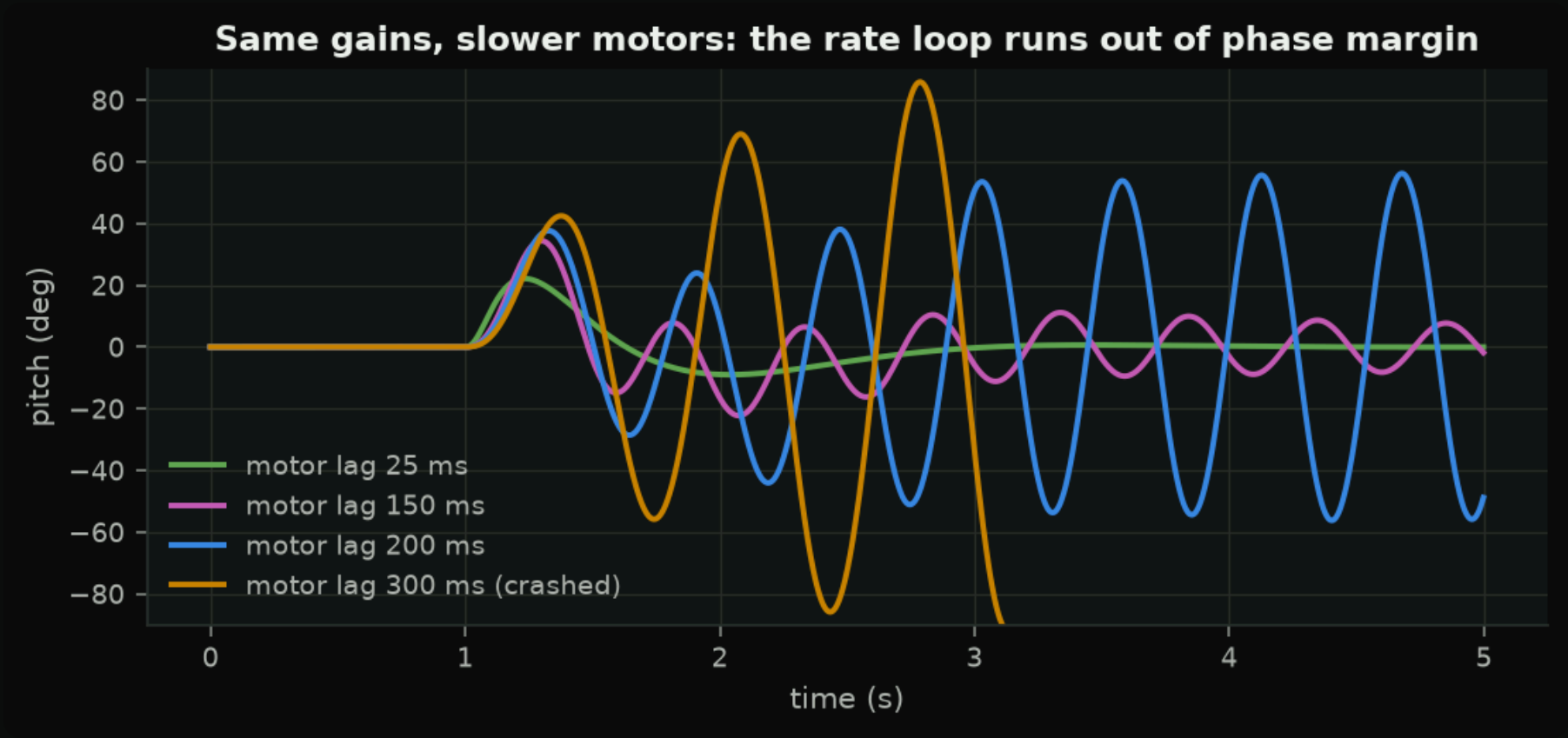
The cascade



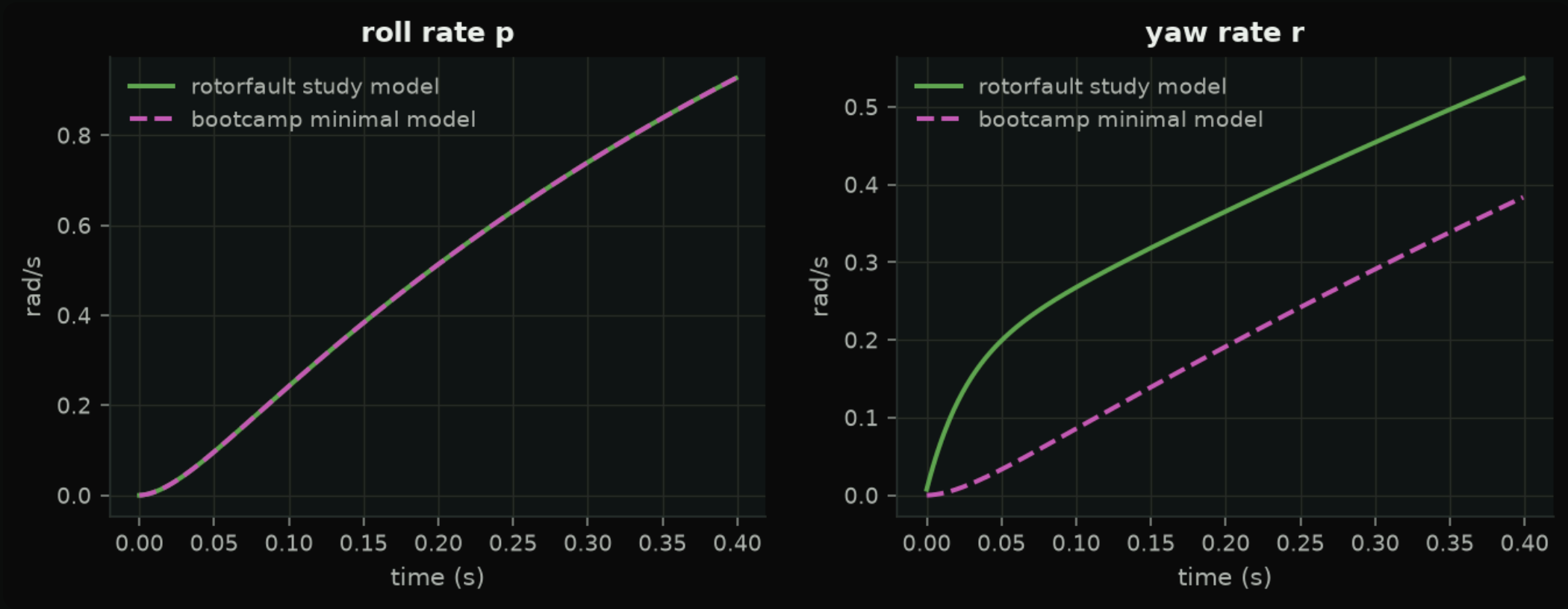
position (P) → velocity (PI) → attitude (P) → rate (PID): each loop several times faster than the one around it.

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Actuator lag sets the inner-loop limit



Minimal model vs research model



Roll agrees to 10^{-5}; yaw does not. What is missing?

Lab 5 · the failure catalogue

1. baseline course
2. slow motors (`motor_scale`)
3. heavier than the controller thinks
4. noisy gyro and the D filter
5. less thrust margin (T/W 1.65) → session 6
6. the rotorfault CLI

Hand in: one plot and two sentences per failure.

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